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Fortuna Brand Health Tracker

Product Documentation — How It Works · Data Requirements · Dependencies

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1. Product Overview

Fortuna is a **CBM-compliant brand health tracking platform** that measures how strongly a brand is present in category buyers' minds at the moments they enter the category. It is designed for brand strategists, insights teams, and marketing planners.

Unlike traditional brand tracking (awareness scores, likeability), Fortuna measures **mental availability** — the probability that a buyer will think of your brand when the need to buy arises — using the **Category Buyer Memory (CBM)** methodology.

> **The Core Question Fortuna Answers:**
> "Which brand do buyers think of — and in which buying situations — when they enter this category?"

Who Is It For?

User	Primary Use
Brand Managers	Track MPen, MMS, and competitive position wave over wave
Insights / Research Teams	Design CBM-compliant surveys, upload fieldwork data, review compliance
Media Planners	Measure effective reach and correct branding for each campaign creative
Strategy Consultants	Use forecast tools and Mental Advantage maps to build brand growth briefs

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2. How the App Works

Current State (v0.5 — UI Build)

Fortuna v0.5 is a **fully functional front-end prototype**. All dashboards, charts, and analytics pages are built and running with **sample/demo data** pre-populated. No backend database or authentication is required to run it locally.

> **Important:** The current version uses hard-coded sample data (Indonesian Credit Cards category — BCA as focal brand). A backend database (Phase 1) is planned but not yet implemented. All data entry via modals is UI-only — it does not persist between page refreshes.

Architecture at a Glance

Layer	What It Does
Next.js App Router	Handles routing, server/client components, page rendering
React Context (SegmentContext)	Shares the active segment (Non-buyers+Light / Heavy / Overall) across all dashboard pages globally

Page Components	Each analytics page is a self-contained React component with its own data constants and chart logic
Recharts	Renders all charts (bar, line, scatter)
DataEntryModal	Reusable component for manual data entry + Excel upload on every analytics page
CBM Engine (lib/)	Frozen formula contracts and compliance validators — not yet wired to live data

User Flow

1. User lands on **http://localhost:3000** — sees the landing page
2. Clicks **Open Dashboard** → enters the dashboard at `/dashboard`
3. The **Segment Toggle** (top right of each page) switches between:
   — *Non-buyers + Light* (default — recruitment focus)
   — *Heavy Buyers* (retention focus)
   — *Overall* (competitive benchmarking)
4. All charts, KPI cards, and analysis text update dynamically when the segment changes
5. Each analytics page has an **Add Data** button — opens a modal for manual entry or Excel upload
6. The **Sidebar** provides navigation across all 11 pages

Segment System

The segment toggle is the most important control in the app. It drives different views across all pages:

Segment	Who	Primary KPI	Strategic Goal
Non-buyers + Light (Default)	Infrequent buyers / never bought	Mental Penetration (MPen)	Recruitment / Growth
Heavy Buyers	Frequent purchasers of your brand	Share of Mind (SoM)	Retention

Overall	Full sample — all category buyers	Mental Market Share (MMS)	Competitive benchmarking

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3. Data Required to Run the App

3.1 To Run Right Now (Demo Mode)

> **No data input required.** The app runs immediately after `npm install && npm run dev`. All pages are pre-populated with sample data for the Indonesian Credit Cards category (BCA as focal brand, Q4 2024 wave).

3.2 To Use With Real Data (Current Version)

Because there is no database yet, real data must be entered directly into the page source files (`app/dashboard/[page]/page.tsx`). Each page has data constants at the top that can be replaced with actual wave values.

3.3 Data Required Per Page

Page	Data Needed	Format
Dashboard Overview	MPen %, MMS %, Network Score, SoM/Prompted Awareness per segment per wave. Competitive analysis text.	% values per brand per segment per wave
Brand KPIs (Analytics)	MPen, MMS, NS, SoM for all brands in category. Both non-light and heavy buyer cuts.	% values per brand per segment
Mental Advantage	Binary linkage matrix: for each brand × CEP combination, the % of buyers who link them. All brands, all CEPs.	Matrix of % values (brands as rows, CEPs as columns)
Physical Availability — Presence	Channel coverage %: for each distribution channel (branches, ATMs, online, agent banking, etc.), what % coverage does the brand have vs category norm?	% per channel, brand vs norm

Physical Availability — Portfolio	SKU-level sales %, penetration %, efficiency ratio for each card product. Duplication of purchase rates between SKUs.	% per SKU
Physical Availability — Prominence	Fame score (% who recognise the asset) and Uniqueness score (% who attribute it only to your brand) for each distinctive asset.	Scores 0–100 per asset
Marketing Reach	Noticed % (effective reach) and Correctly Branded % per creative execution per wave. Tested on de-branded stimuli.	% per creative per wave
WOM Tracker	PWOM and NWOM per 100 category buyers for each brand per wave. Binary measurement (yes/no in past month).	Count per 100 buyers per brand per wave
Forecast — DJ Forecast	Penetration % and purchase frequency per brand per wave. At least 3 brands to fit the DJ curve.	% pen + frequency (times/year) per brand
Forecast — MMS → Sales	MMS % and SMS (Sales Market Share %) per wave, matched quarter.	% per wave (minimum 4 data points to calculate r)
CEP Builder	List of Category Entry Points for the category (buying occasions). Categorised as CEP / Baseline Competency / Secondary Attribute.	Text list with type classification
Surveys	Fieldwork data: respondent-level or aggregate binary linkage scores per brand × CEP. Exported from your survey platform (SPSS, Excel, etc.).	Excel file matching the downloadable template format

3.4 Sample Survey Data Format

When uploading via the Data Entry Modal, the expected CSV template columns are:

Page	CSV Template Headers

Mental Advantage	Brand, CEP, Actual %, Expected %
Physical Availability — Presence	Channel, BCA Coverage %, Category Norm %
Physical Availability — Portfolio	SKU, Sales %, Penetration %, Efficiency Ratio
Physical Availability — Prominence	Asset, Fame Score, Uniqueness Score
Marketing Reach	Creative Name, Noticed %, Correctly Branded %, Wave
WOM Tracker	Brand, PWOM, NWOM, Wave
DJ Forecast	Wave, Brand, Penetration %, Frequency (times/yr)
MMS → Sales	Wave, MMS %, SMS %

> **How to download templates:** In any page, click the **Add Data** button (top right). In the modal that opens, click **Download Template**. A pre-formatted CSV file will be saved to your downloads folder.

3.5 Data Quality Requirements (CBM Methodology)

Requirement	Why It Matters
Survey format: binary pick-any only	Likert scales introduce acquiescence bias and inflate all brands equally. Binary forces genuine top-of-mind association.
Brand ordering: alphabetical	Position bias — brands listed first get higher selection rates. Alphabetical order eliminates this.
No superlatives in CEP wording	"Most trusted" biases toward market leaders. Use plain-form: "trusted."

Non-buyer quota ≥30% of sample	Non-buyers are the growth pool. Under-sampling them skews all growth metrics.
Minimum n=500 category buyers per wave	Below this, estimates are unstable. For subgroups, n≥100 required.
Same sample base for all brands	Scores must be comparable — different bases cannot be fairly benchmarked.
CEP measured in past 12 months screener	Bounded recall ensures only active category buyers are included.

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4. Page-by-Page Guide

4.1 Dashboard Overview `/dashboard`

Segment-aware command centre. Shows 4 KPI cards, a segment analysis panel, and a MMS vs SMS gap diagnosis. The content changes entirely when you switch segments.

- **KPI Cards:** MPen, MMS, NS, SoM (or Prompted Awareness) — each with wave-on-wave delta
- **Segment Banner:** Active segment + primary KPI + strategic goal label
- **Key Insight:** CEO-ready headline + recommendation — unique per segment
- **MMS vs SMS Gap:** Diagnoses whether the problem is mental (MMS < SMS) or physical (MMS > SMS)

4.2 Brand KPIs `/dashboard/analytics`

Competitive benchmarking. All brands ranked by the focal segment's primary KPI. DJ-normalized bar chart.

4.3 Mental Advantage `/dashboard/mental-advantage`

Brand × CEP heatmap. Each cell shows deviation from DJ-expected score.

Cell Label	Deviation	Action

DEFEND	≥+5pp above expected	Protect in media — this is a mental asset
MAINTAIN	Within ±4pp	Hold activity — no urgent action
BUILD	≤−5pp below expected	Target in next campaign — mental gap

4.4 Physical Availability `/dashboard/physical`

Five tabs: Overview · Presence (channel coverage) · Portfolio (SKU contribution) · Prominence (distinctive assets) · Gap Analysis (MMS vs SMS diagnosis per segment).

4.5 Marketing Reach `/dashboard/reach`

Creative effectiveness tracker. Noticed % + Correct Branding % + Branding Ratio per creative per wave. Below 60% Branding Ratio = "vampire creative" flag.

4.6 WOM Tracker `/dashboard/wom`

PWOM and NWOM per 100 category buyers. Net WOM = PWOM − NWOM. WOM Ratio = PWOM ÷ NWOM (>4× is strong).

4.7 Forecast `/dashboard/forecast`

Five forecast methodologies in tabs:

Tab	What It Does
DJ Forecast	Scatter chart of all brands on the Double Jeopardy curve. Scenario builder: set target penetration → predicted frequency output.
CEP Expected Scores	Full brand × CEP matrix with Actual / Expected / Deviation — numerical version of the heatmap.
MMS → Sales	Dual-line historical chart + projection. MMS leads SMS by 1–2 quarters at r=0.83.
AI Predictions	Media spend multiplier slider → projected MPen & MMS. Category twin benchmarks. Trend alerts.

Growth Potential	MPen by segment, growth waterfall, recruitment vs network size priority.

4.8 CEP Builder `/dashboard/ceps`

Build a CBM-compliant attribute list using the 7W framework. Composition meter enforces 60–70% CEPs. CBM Engine flags wording violations in real time.

4.9 Surveys `/dashboard/surveys`

Block-based survey canvas. 17 question blocks in the library. Core blocks (mandatory) cannot be removed. Upload Results tab accepts Excel fieldwork data.

4.10 Reports `/dashboard/reports`

Generate: Executive Summary PDF · Full Deck PPTX · CBM Compliance Report · Raw Data XLSX.

4.11 Help `/dashboard/help`

Full in-app product manual — covers every page, how to use it, how to interpret the data, methodology guide, KPI reference (8 metrics), and FAQ.

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5. CBM Methodology Reference

The CBM Mantra

#	Principle	What It Means in Practice
1	Design for the Category	CEPs belong to the category, not the brand. Every brand is measured against the same buying occasions.
2	Analyse for the Buyer	Non-buyers + light buyers are the default, primary segment. They are the growth pool.
3	Report for the Brand	Scores are DJ-normalized — small and large brands are fairly compared by deviation from expected.

Double Jeopardy (DJ) Law

Brands with lower penetration suffer *double* — they have fewer buyers AND those buyers buy them less frequently. This is a market law, not a strategy. Brands naturally cluster along the DJ curve.

- **Above the DJ curve:** structural advantage (unusual)
- **On the DJ curve:** performance matches brand size
- **Below the DJ curve:** loyalty problem — buyers leaving faster than expected

Category Entry Points (CEPs)

A CEP is a situational trigger that activates category buying. CEPs are buying occasions — not brand attributes. They are derived using the **7W Framework**:

W	Question	Example CEP (Credit Cards)
Who	Who is the buyer?	"When paying for my family's expenses"
What	What is the occasion?	"When making an online purchase"
When	When does the need arise?	"When travelling internationally"
Where	Where does it happen?	"When checking out at a restaurant"
Why	What triggers it?	"When I need emergency cash"
With What	What tool is used?	"When tapping to pay contactlessly"
With Whom	Who is involved?	"When splitting a bill with friends"

MMS vs SMS Gap — Diagnosis Framework

Condition	Meaning	Recommended Action

MMS > SMS (positive gap)	Physical availability problem — buyers think of the brand but can't find or buy it	Expand distribution. Close channel coverage gaps.
MMS < SMS (negative gap)	Mental availability problem — sales driven by habit, mental links are fragile	Reinforce CEP coverage in media. Build distinctive assets.
MMS ≈ SMS (gap < ±3pp)	Healthy parity	Maintain current investment mix.

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6. Dependencies & Tech Stack

6.1 Runtime Requirements

Requirement	Version	Where to Download
Node.js	18.0 or higher	https://nodejs.org
npm	Comes with Node.js	Included with Node
Git	Any recent version	https://git-scm.com

> **That's it for the current UI build.** No database, no Redis, no Python service, no API keys required. Just Node.js 18+ and npm.

6.2 Core npm Dependencies

Package	Version	Purpose

next	14.2.35	React framework — App Router, server/client components, routing
react	^18	UI component library
react-dom	^18	React DOM renderer
typescript	^5	TypeScript language support
tailwindcss	^3.4.1	Utility-first CSS framework — all styling
recharts	^2.15.3	Chart library — all bar, line, scatter charts
lucide-react	^0.513.0	Icon library — all icons throughout the UI
clsx	^2.1.1	Conditional className utility
tailwind-merge	^3.3.0	Merges conflicting Tailwind classes
class-variance-authority	^0.7.1	Variant-based component styling (used in Button)

6.3 Dev Dependencies

Package	Purpose
@types/node	TypeScript types for Node.js
@types/react	TypeScript types for React

@types/react-dom	TypeScript types for React DOM
eslint	JavaScript/TypeScript linter
eslint-config-next	Next.js ESLint ruleset
postcss	CSS post-processor (required by Tailwind)
autoprefixer	Adds CSS vendor prefixes automatically

6.4 Planned Dependencies (Phase 1+, Not Yet Installed)

Package	Purpose	Phase
@prisma/client + prisma	ORM — database models and queries	Phase 1
@clerk/nextjs	Authentication and organisation management	Phase 1
@trpc/server + @trpc/client	Type-safe API layer	Phase 1
@anthropic-ai/sdk	Claude API — AI CEP generation, trend analysis	Phase 5
resend	Transactional email (survey invites, reports)	Phase 6
ioredis	Redis client for caching analytics results	Phase 3

6.5 Full Tech Stack (Planned Architecture)

Layer	Technology
Frontend	Next.js 14, TypeScript, Tailwind CSS, Recharts
Authentication	Clerk (org-based multi-tenancy)
API	tRPC (type-safe, end-to-end)
Database ORM	Prisma
Database	PostgreSQL (hosted on Supabase)
Analytics Engine	Python FastAPI microservice (DJ normalization at scale)
Caching	Redis
AI	Anthropic Claude API (CEP generator, category twin matching)
Email	Resend
CI/CD	GitHub Actions → Vercel (frontend)
Testing	Vitest (unit), Playwright (e2e)

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7. Setup & Running Locally

Step 1 — Install Node.js

Download and install Node.js 18 or higher from **https://nodejs.org**. The LTS version is recommended.

Verify after installation:

```
node --version   # should show v18.x.x or higher

npm --version    # should show 9.x.x or higher
```

Step 2 — Clone the Repository

```
git clone https://github.com/matiyashu/Brand-Health-Tracker---Mental-Availability-Physical-Availability.git

cd Brand-Health-Tracker---Mental-Availability-Physical-Availability
```

Step 3 — Install Dependencies

```
npm install
```

This reads `package.json` and installs all required packages into the `node_modules/` folder. Takes 30–60 seconds on first run.

Step 4 — Run the Development Server

```
npm run dev
```

Open your browser and go to: **http://localhost:3000**

> **No environment variables needed.** The app runs in demo mode with sample data. No database connection string, no API keys, no .env file required for the current UI build.

Step 5 — Build for Production (Optional)

```
npm run build   # creates optimised production build

npm start       # serves the production build
```

Available npm Scripts

Command	What It Does
`npm run dev`	Start development server with hot reload (port 3000)
`npm run build`	Create optimised production build (also runs ESLint + TypeScript checks)

`npm start`	Serve the production build
`npm run lint`	Run ESLint across all TypeScript/TSX files

Folder Structure

```
brandhealth/
├── app/                          Next.js pages (App Router)
│   ├── page.tsx                  Landing page
│   ├── layout.tsx                Root layout
│   ├── globals.css               Global styles + colour tokens
│   └── dashboard/
│       ├── layout.tsx            Dashboard shell + SegmentProvider
│       ├── page.tsx              Dashboard overview
│       ├── analytics/            Brand KPIs
│       ├── mental-advantage/     Mental Advantage heatmap
│       ├── physical/             Physical Availability (5 tabs)
│       ├── reach/                Marketing Reach
│       ├── wom/                  WOM Tracker
│       ├── forecast/             Forecast Engine (5 tabs)
│       ├── ceps/                 CEP Builder
│       ├── surveys/              Survey builder + upload
│       ├── reports/              Report generation
│       ├── help/                 Product manual
│       └── settings/             Settings
├── components/                   Shared React components
│   ├── sidebar.tsx               Navigation
│   ├── segment-toggle.tsx        Segment switcher
│   ├── data-entry-modal.tsx      Manual entry + Excel upload modal
```

```
│   └── ...
├── contexts/
│   └── segment-context.tsx       Global segment state
├── lib/
│   ├── utils.ts                  cn() class utility
│   └── cbm-engine/               CBM formula contracts + validators
│       ├── formula-contracts/    FROZEN — MPen, MMS, NS, SoM formulas
│       ├── mantra-validators/    CBM compliance checks
│       ├── attribute-rules/      CEP wording linter
│       └── survey-guards/        Pre-launch survey checklist
├── package.json                  Dependencies manifest
├── tailwind.config.ts            Tailwind configuration + custom colours
├── tsconfig.json                 TypeScript configuration
└── next.config.mjs               Next.js configuration
```

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8. Formula Reference

KPI	Formula	Target / Benchmark
Mental Penetration (MPen)	Buyers linking ≥1 CEP to brand ÷ Total sample × 100	>50% is strong for a market leader
Mental Market Share (MMS)	Brand's total CEP links ÷ Total CEP links (all brands) × 100	Should track close to Sales Market Share
Network Score (NS)	Total CEP links for brand ÷ Buyers with MPen for that brand	>4.0 is healthy breadth

Share of Mind (SoM)	Brand CEP links ÷ Total links among MPen base × 100	Benchmark against MMS
DJ Expected Score	(Row Total × Column Total) ÷ Grand Total	Used to calculate Mental Advantage deviation
Mental Advantage Deviation	Actual Score − DJ Expected Score	≥+5pp = DEFEND · ±4pp = MAINTAIN · ≤−5pp = BUILD
Net WOM	PWOM − NWOM	Positive is healthy; negative is urgent
WOM Ratio	PWOM ÷ NWOM	>4× is strong advocacy balance
Branding Ratio (Reach)	Correctly Branded % ÷ Noticed %	>70% is strong; <60% = "vampire creative"
MMS → SMS Correlation	r = 0.83 (MMS leads SMS by 1–2 quarters)	MMS > SMS = physical gap; SMS > MMS = mental gap

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